

$T^{\mu}_{\nu} = \begin{pmatrix} -\rho & 0 & 0 & 0 \\ 0 & p & 0 & 0 \\ 0 & 0 & p & 0 \\ 0 & 0 & 0 & p \end{pmatrix}$

 (Annotations: ρ = energy density, p = pressure)

$P = w \rho$

 $w = 0$ "matter" $\sim a^{-3}$

 $w = \frac{1}{3}$ radiation $\sim a^{-4}$

 $w = -1$ C.C. $\sim \text{const}$

$ds^2 = -dt^2 + a^2(t) \left[\frac{dr^2}{1-kr^2} + r^2(d\theta^2 + \sin^2\theta d\phi^2) \right]$

 (Annotations: Robertson-Walker, scale factor $K = \begin{cases} 0 & \text{"flat"} \\ 1 & \text{"closed"} \\ -1 & \text{"open"} \end{cases}$)

$$g^{\mu\nu} \left(R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R \right) = 8\pi G_N T^{\mu}_{\nu}$$

$$R - \frac{d}{dt} R = 8\pi G_N T^{\mu}_{\mu}$$

$R = 8\pi G_N T^{\mu}_{\mu} = 8\pi G_N (-\rho + 3p)$

 $R_{tt} = -3\frac{\ddot{a}}{a}$

 Plug into Einstein's Eqn:

$$\left(\ddot{a} + \frac{1}{2} g_{tt} R \right) = 8\pi G_N T_{tt}$$

$$-\frac{3\ddot{a}}{a} + \frac{1}{2} 8\pi G_N (\rho - 3p) = 8\pi G_N \rho$$

$$R_{00} = n^2 [2(K + \dot{a}^2) + a\ddot{a}]$$

$$T_{00} = g_{00} T^0_0 = n^2 a^2 p$$

$$\frac{\ddot{a}}{a} = -\frac{4\pi}{3} G_N (\rho + 3p) \quad (F2)$$

$$\Rightarrow R_{00} - \frac{1}{2} g_{00} R = 8\pi G_N T_{00}$$

$$\hookrightarrow n^2 [2(K + \dot{a}^2) + a\ddot{a}] - \frac{1}{2} n^2 a^2 8\pi G_N (\rho - 3p) = 8\pi G_N \frac{n^2 a^2 p}{1}$$

$$\hookrightarrow 2(K + \dot{a}^2) + a(-a \cdot \frac{4\pi}{3} G_N (\rho + 3p)) = \frac{8\pi}{2} G_N a^2 (2p + \rho - 3p)$$

$$\hookrightarrow 2 \left[\frac{K}{a^2} + \left(\frac{\dot{a}}{a} \right)^2 \right] = 4\pi G_N (\rho - \cancel{p} + \frac{\rho}{3} \cancel{+ p})$$

$$\hookrightarrow \left(\frac{\dot{a}}{a} \right)^2 = \frac{8\pi}{3} G_N \rho - \frac{K}{a^2} \quad (F1)$$

$$H_0^2 \equiv \frac{8\pi}{3} G_N \rho_{\text{critical}}$$

$$\text{If } \rho_{\text{observed}} = \rho_{\text{critical}} \Rightarrow \frac{K}{a^2} = 0$$

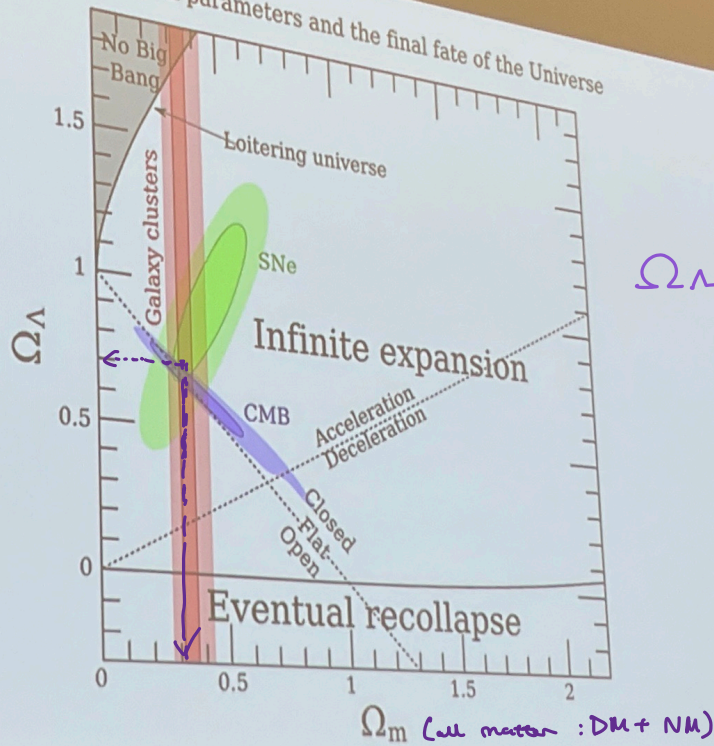
$\hookrightarrow$ universe must be flat.

$$\rho_{\text{obs}} > \rho_{\text{crit}} \Rightarrow K = +1$$

$$\rho_{\text{obs}} < \rho_{\text{crit}} \Rightarrow K = -1$$

$$H_0 = 69.3 \pm 0.8 \frac{\text{km/s}}{\text{Mpc}} \rightarrow 1 \text{ Mpc} = 3.26 \text{ million light years.}$$

Cosmological parameters and the final fate of the Universe



$$\Omega_\Lambda = \frac{\rho_\Lambda}{\rho_{critical}}$$

Hubble Tension: 2σ apart.

$K=0 \Rightarrow$ observations

$$H_0^2 \equiv \frac{8\pi}{3} G_N \rho_{\text{critical}}$$

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi}{3} G_N \rho_{\text{critical}} (\underbrace{\Omega_m}_0.3 a^{-3} + \Omega_{r,0} a^{-4} + \underbrace{\Omega_\Lambda}_{\text{cosmological constant}})$$

as you go back in time

go back in a

each term picks up diff. factors of $\frac{1}{2}$.

$$a \rightarrow \frac{a}{2}$$

$$\left(\frac{\dot{a}}{a}\right)^2 \propto a^{-n}$$

$$\left(\frac{\dot{a}}{a}\right) \propto a^{-n/2}$$

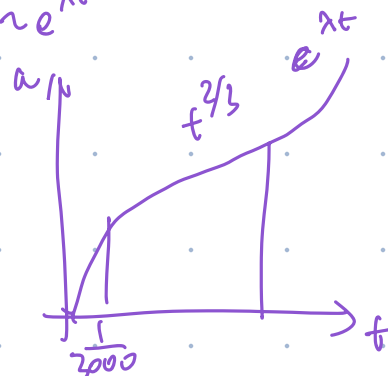
$$da \cdot a^{\frac{n}{2}-1} \propto dt$$

$$t \propto a^{n/2} \Rightarrow a \propto t^{2/n}$$

$$n=4 \quad a \propto t^{1/2}$$

$$n=3 \quad a \propto t^{2/3}$$

$$\text{c.c.} \quad a \sim e^{\lambda t}$$



$$\left(\frac{\dot{a}}{a}\right)^2 = H_0^2 f(a)$$

$$\left(\frac{\dot{a}}{a}\right) = H_0 \sqrt{f(a)}$$

$$\frac{da}{a H_0 \sqrt{f(a)}} = dt$$

$$t_{\text{universe}} = \int_1^{a_{\text{now}}} dt$$

$$= \int_0^1 \frac{da}{a H_0 \sqrt{f(a)}}$$

Matter Domination

$$= \frac{1}{H_0} \int_0^1 \frac{da}{a^2 \sqrt{a^{-3}}}$$

$$= \frac{1}{H_0} \int_0^1 \frac{da}{\Omega_m a^{-1/2}}$$

$$= \frac{1}{H_0} \int_0^1 \frac{da}{\Omega_m} \sqrt{a}$$

$$= \frac{2}{3} \frac{1}{H_0}$$

$$a^{-\frac{3}{2}} \cdot a = a^{-\frac{1}{2}}$$

include
if you ~~mean~~ Ω_m

↓
you get
 $\Omega_m = 0.96$

✓
 $\frac{t}{H_0} = 1.39 \times 10^{10} \text{ years}$